9. Which of the following is not a scalar?
   (a) viscosity
   (b) surface tension
   (c) pressure
   (d) stress

10. If the temperature of the wire is increased, then the Young's modulus will
   (a) remain the same
   (b) decrease
   (c) increase rapidly
   (d) increase by very a small amount

11. Copper of fixed volume V is drawn into a wire of length l. When this wire is subjected to a constant force F, the extension produced in the wire is Δl. If Y represents the Young's modulus, then which of the following graphs is a straight line?
   (NEET 2014 model)
   (a) Δl versus V
   (b) Δl versus Y
   (c) Δl versus F
   (d) Δl versus $\frac{1}{l}$

12. A certain number of spherical drops of a liquid of radius R coalesce to form a single drop of radius R and volume V. If T is the surface tension of the liquid, then
   (a) energy = 4 V T $\left(\frac{1}{r}-\frac{1}{R}\right)$ is released
   (b) energy = 3 V T $\left(\frac{1}{r}+\frac{1}{R}\right)$ is absorbed
   (c) energy = 3 V T $\left(\frac{1}{r}-\frac{1}{R}\right)$ is released
   (d) energy is neither released nor absorbed

13. The following four wires are made of the same material. Which of these will have the largest extension when the same tension is applied?
   (a) length = 200 cm, diameter = 0.5 mm
   (b) length= 200 cm, diameter = 1 mm
   (c) length = 200 cm, diameter = 2 mm
   (d) length= 200 cm, diameter = 3 m

14. The wettability of a surface by a liquid depends primarily on
   (a) viscosity
   (b) surface tension
   (c) density
   (d) angle of contact between the surface and the liquid

15. In a horizontal pipe of non-uniform cross section, water flows with a velocity of 1 m s^{-1} at a point where the diameter of the pipe is 20 cm. The velocity of water (1.5 m s^{-1}) at a point where the diameter of the pipe is (in cm)
   (a) 8 (b) 16
   (c) 24 (d) 32

Answers:

1) c	2) a	3) a	4) d
5) d	6) b	7) c	8) d
9) d	10) b	11) c	12) c
13) a	14) d	15) b	

II. Short Answer Questions

1. Define stress and strain.
2. State Hooke's law of elasticity.
3. Define Poisson's ratio.
4. Explain elasticity using intermolecular forces.
5. Which one of these is more elastic, steel or rubber? Why?

UNIT-7(XI-Physics_Vol-2).indd 91 02-01-2023 14:40:22